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COURSE NAME: ANRENNNA AND WAVE PROPAGATION

EX NO: 8A

SIMULATION OF EMI SOURCES AND INTERFERENCE ANALYSIS USING MATLAB

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OBJECTIVES

- i) To generate and simulate different Electromagnetic Interference (EMI) sources using MATLAB.
 - ii) To analyze the effect of EMI on a desired communication signal in time and frequency domains.
 - iii) To evaluate Signal-to-Noise Ratio (SNR) degradation and study interference mitigation techniques using MATLAB.
-

OBJECTIVE 1

To Generate and Simulate EMI Sources Using MATLAB

PROCEDURE

1. Open MATLAB and create a new script file.
2. Define the sampling frequency and simulation time.
3. Generate the desired communication signal.
4. Generate EMI sources such as:
 - Sinusoidal interference
 - Impulse noise
 - Broadband noise
5. Combine the desired signal with EMI sources.
6. Plot the generated waveforms.
7. Observe the interference characteristics.

FORMULA :

Desired Signal:

$$s(t) = A \sin(2\pi f t)$$

EMI Signal:

$$i(t) = B \sin(2\pi f_i t)$$

Received Signal:

$$r(t) = s(t) + i(t)$$

PROGRAM (MATLAB CODE)

```
clc;
```

```
clear;
```

```
close all;
```

```
fs = 10000;
```

```
t = 0:1/fs:0.01;
```

```
signal = sin(2*pi*1000*t);
```

```
emi = 0.5*sin(2*pi*2500*t);
```

```
received = signal + emi;
```

```
figure;
```

```
subplot(3,1,1);
```

```
plot(t,signal);
```

```
title('Desired Signal');
```

```
subplot(3,1,2);
```

```
plot(t,emi);
```

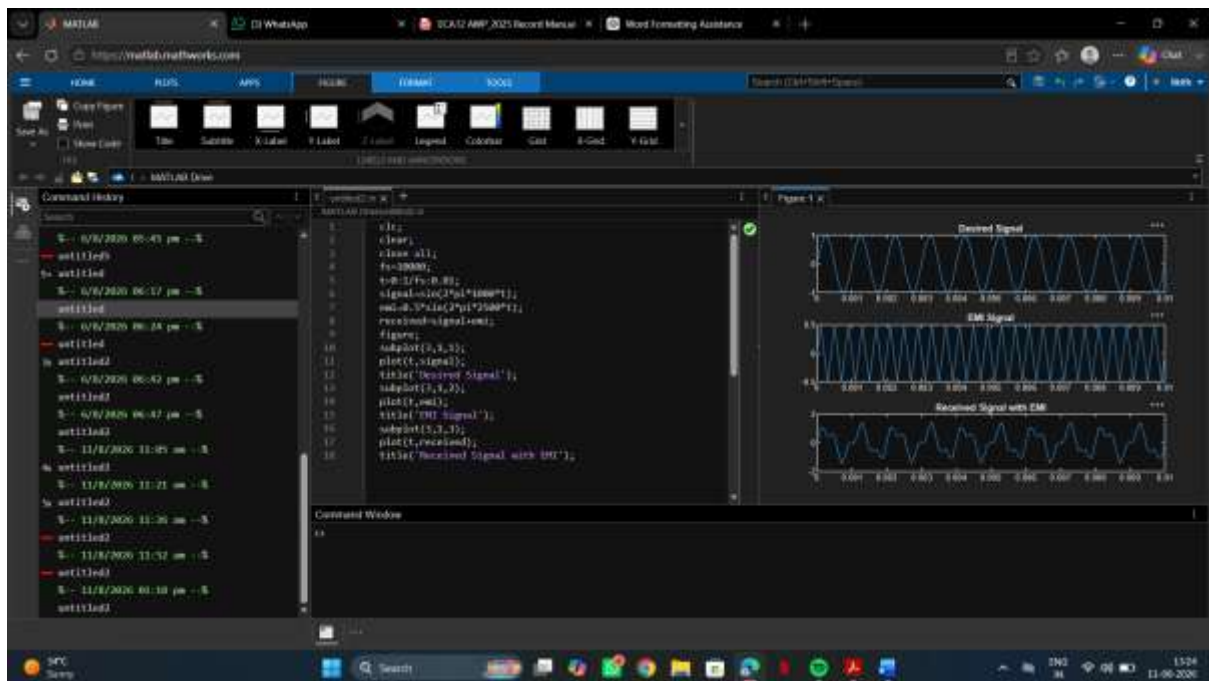
```
title('EMI Signal');
```

```
subplot(3,1,3);
```

```
plot(t,received);
```

```
title('Received Signal with EMI');
```

MODEL GRAPH



RESULT

Various EMI sources were successfully generated and simulated using MATLAB.

OBJECTIVE 2

To Analyze EMI Effects in Time and Frequency Domain

PROCEDURE

1. Generate the desired signal.
2. Introduce EMI into the signal and apply FFT to obtain the frequency spectrum.
3. Observe spectral components and identify interference frequencies.
4. Compare the original and corrupted signals.

PROGRAM (MATLAB CODE)

```
N = length(received);
```

```
f = (-N/2:N/2-1)*(fs/N);
```

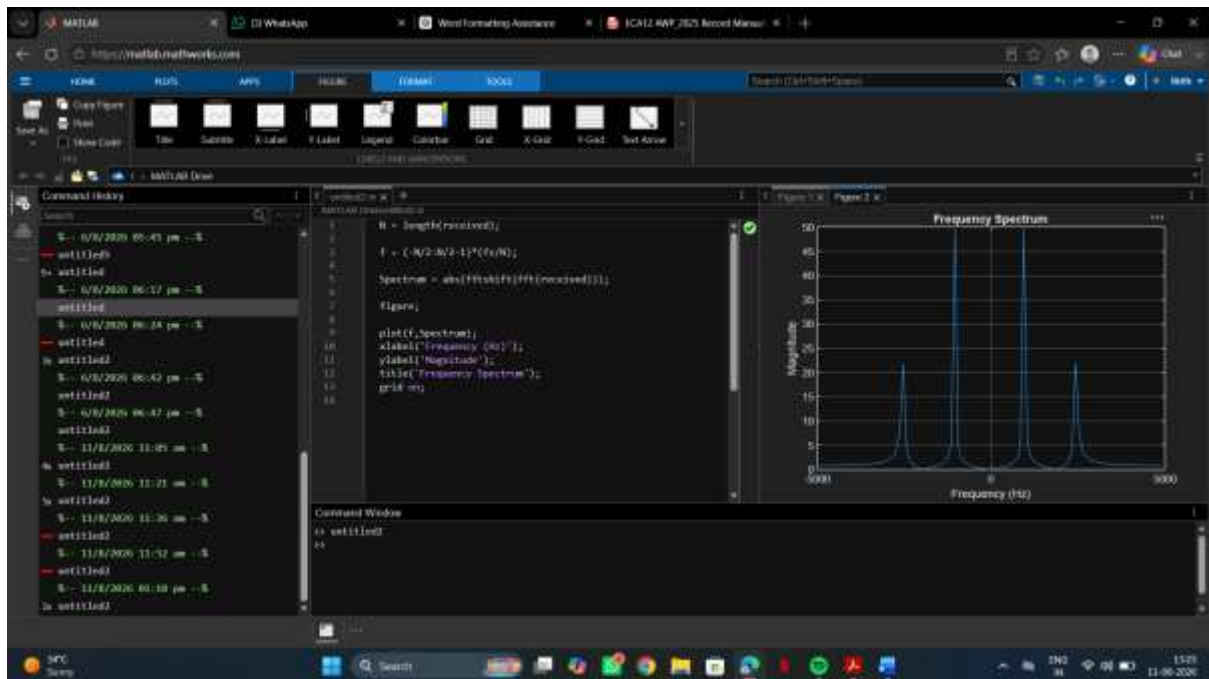
```
Spectrum = abs(fftshift(fft(received)));
```

```
figure;
```

```
plot(f,Spectrum);
```

```
xlabel('Frequency (Hz)');  
ylabel('Magnitude');  
title('Frequency Spectrum');  
grid on;
```

MODEL GRAPH



RESULT

The interference frequencies were successfully identified in the frequency domain using FFT analysis.

OBJECTIVE 3

To Evaluate SNR Degradation and EMI Mitigation

PROCEDURE

1. Calculate signal power.
2. Calculate interference power.
3. Determine SNR.
4. Apply filtering techniques.
5. Recalculate SNR after filtering.
6. Compare system performance before and after mitigation.

FORMULA

$$\text{SNR} = 10 \log_{10}(P_s / P_n)$$

PROGRAM (MATLAB CODE)

```
Ps = mean(signal.^2);
Pn = mean(emi.^2);
SNR_before = 10*log10(Ps/Pn);
filtered = lowpass(received,1500,fs);
Pn_after = mean((filtered-signal).^2);
SNR_after = 10*log10(Ps/Pn_after);
fprintf('SNR Before = %.2f dB\n',SNR_before);
fprintf('SNR After = %.2f dB\n',SNR_after);
```

MODEL GRAPH

RESULT

The SNR degradation caused by EMI was analyzed and improved using filtering techniques in MATLAB.

OVERALL MATLAB PROGRAM

```
clc;
clear;
close all;

%% Objective 1: Generate Desired Signal and EMI Sources

fs = 10000;           % Sampling Frequency
t = 0:1/fs:0.01;      % Time Vector

% Desired Communication Signal
A1 = 1;
f1 = 1000;
signal = A1*sin(2*pi*f1*t);

% EMI Source 1 (Sinusoidal Interference)
Aemi = 0.5;
```

```

femi = 2500;
emi_sine = Aemi*sin(2*pi*femi*t);
% EMI Source 2 (Broadband Noise)
emi_noise = 0.3*randn(size(t));
% EMI Source 3 (Impulse Noise)
emi_impulse = zeros(size(t));
emi_impulse(20:20:end) = 1;
% Total EMI
emi_total = emi_sine + emi_noise + emi_impulse;
% Received Signal
received = signal + emi_total;

```

Objective 2:

Time Domain and Frequency Domain Analysis :

```

N = length(received);
f = (-N/2:N/2-1)*(fs/N);
Signal_FFT = abs(fftshift(fft(signal)))/N;
EMI_FFT = abs(fftshift(fft(emi_total)))/N;
Received_FFT = abs(fftshift(fft(received)))/N;
%% Objective 3: SNR Analysis and EMI Mitigation
% Signal Power
Ps = mean(signal.^2);
% Noise Power Before Filtering
Pn_before = mean(emi_total.^2);
% SNR Before Filtering
SNR_before = 10*log10(Ps/Pn_before);
% Low Pass Filter
filtered = lowpass(received,1500,fs);
% Noise Power After Filtering
Pn_after = mean((filtered-signal).^2);
% SNR After Filtering

```

```

SNR_after = 10*log10(Ps/Pn_after);

%% Display Results

fprintf('\n-----\n');
fprintf('    EMI ANALYSIS RESULTS\n');
fprintf('-----\n');
fprintf('Signal Frequency = %d Hz\n',f1);
fprintf('EMI Frequency = %d Hz\n',femi);
fprintf('\nSignal Power = %.4f W\n',Ps);
fprintf('Noise Power Before = %.4f W\n',Pn_before);
fprintf('Noise Power After = %.4f W\n',Pn_after);
fprintf('\nSNR Before Filtering = %.2f dB\n',SNR_before);
fprintf('SNR After Filtering = %.2f dB\n',SNR_after);
fprintf('SNR Improvement = %.2f dB\n', ...
    (SNR_after-SNR_before));

```

```

%% Plot Results

figure('Name','EMI Simulation and Analysis', ...
    'NumberTitle','off');

subplot(4,2,1);
plot(t,signal);
grid on;
title('Desired Signal');
xlabel('Time (s)');
ylabel('Amplitude');

subplot(4,2,2);
plot(t,emi_sine);
grid on;
title('Sinusoidal EMI');
xlabel('Time (s)');
ylabel('Amplitude');

subplot(4,2,3);

```

```
plot(t,emi_noise);
grid on;
title('Broadband EMI Noise');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(4,2,4);
stem(t,emi_impulse);
grid on;
title('Impulse EMI');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(4,2,5);
plot(t,received);
grid on;
title('Received Signal with EMI');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(4,2,6);
plot(f,Received_FFT);
grid on;
title('Frequency Spectrum of EMI Signal');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
subplot(4,2,7);
plot(t,filtered);
grid on;
title('Filtered Signal');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(4,2,8);
bar([SNR_before SNR_after]);
```


grid on;

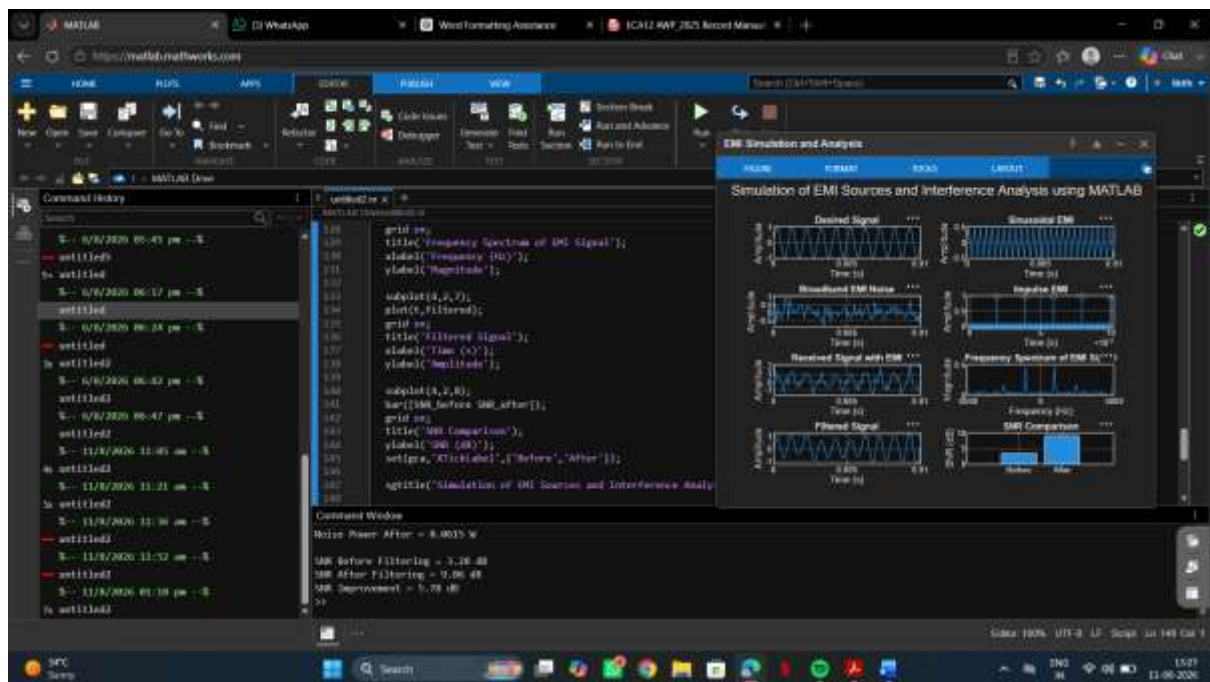
title('SNR Comparison');

ylabel('SNR (dB)');

set(gca,'XTickLabel',{'Before','After'});

sgtitle('Simulation of EMI Sources and Interference Analysis using MATLAB');

OUTPUT



FINAL RESULT

The EMI sources were successfully generated and simulated using MATLAB. The effect of electromagnetic interference on the desired communication signal was analyzed in both time and frequency domains. The interference frequencies were identified using FFT analysis, and the SNR degradation before and after filtering was evaluated. The results demonstrated that filtering techniques can improve the signal quality and reduce the effects of EMI.